

Decathlon Sales & Customer Analytics

Project Objective:

The main objective of this project is to analyze the retail sales data to gain actionable insight that will enhance the performance of the company.

Business Problem Statement:

- Calculate key business metrics such as Total Sales, Total Orders, Total Customers, and Average Order Value.
- Analyze yearly sales performance.
- Compare sales distribution between male and female customers.
- Identify top five sports categories generating the highest sales revenue per month.
- Measure customer retention by identifying repeat customers and show overall retention rate in a year

Data Cleaning & Transformation:

Power Query was used to prepare the dataset before analysis.

- Checking missing values.
- Standardizing data types.
- Formatting dates.
- Creating Month column.
- Creating Year column.
- Removing unnecessary columns.

Data Modeling & DAX Measure:

The cleaned dataset was loaded into Excel's Data Model. Creating another table as calendar using power pivot to establish relationship between calendar and table1 required for enabling efficient DAX calculations and dashboard reporting.

Several business KPIs were created using DAX such as average order value, previous year sales, repeat customer, customer retention rate etc.

Dashboard Development:

The dashboard was designed with a Business Intelligence approach it includes:

- KPI Cards

- Sales Trend
- Top 5 Sports
- Product Category Analysis
- Customer Retention
- Gender Distribution

Interactive slicers allow filtering by monthly and yearly hence dashboard components update dynamically based on user selections.

Key Business Insight:

The dashboard highlights several important findings:

- Total Sales exceeded ₹36 Crores.
- Approximately 30,000 orders were analyzed.
- Around 9,000 unique customers contributed to sales.
- Average Order Value exceeded ₹12,000.
- Customer Retention remained close to 65%.
- Product category contribution varies significantly across sports.
- Monthly filtering reveals seasonal demand for different sports.

Business Impact:

The dashboard provides business users with a centralized reporting solution capable of:

- Monitoring business performance.
- Tracking customer behavior.
- Identifying top-performing products.
- Understanding seasonal sales trends.
- Supporting data-driven decision-making.

Skill Demonstrated:

- Microsoft Excel
- Power Query
- Power Pivot
- DAX
- Data Modeling
- Pivot Tables
- Pivot Charts
- KPI Design
- Business Intelligence
- Dashboard Design
- Data Visualization

- Data Cleaning
- Deployment

Challenges Faced:

- Cleaning inconsistent transaction data.
- Designing a responsive dashboard layout.
- Creating reusable DAX measures.
- Connecting GitHub-hosted images with Excel's IMAGE() function.
- Managing PivotTable sorting for dynamic sports rankings.
- Synchronizing slicers across multiple PivotTables.
- Optimizing dashboard readability without overcrowding the interface.

Future Improvements:

- Integrating live sales data.
- Migrating the dashboard to Power BI.
- Adding regional sales analysis.
- Forecasting future sales using predictive models.
- Building customer segmentation dashboards.

Conclusion:

This project demonstrates how Microsoft Excel can be used as a complete Business Intelligence platform when combined with Power Query, Power Pivot, DAX, and interactive dashboard reporting.